

Building a Career in AI - Field Notes

An original summary of widely-known guidance on starting a career in artificial intelligence, written for this book's examples.

Who shaped this advice

Much of the popular guidance on entering AI comes from Andrew Ng, the founder of DeepLearning.AI and a long-time educator in the field. He is well known for the line that "AI is the new electricity," meaning that, like electricity a century ago, AI is a general-purpose technology that will transform and improve nearly every area of human life and industry.

Three steps to career growth

A simple framing for building a career in AI is three steps: learning foundational skills, working on projects, and finding a job. All three are easier when you are part of a supportive community of peers and mentors. These three steps also map to a natural progression: first you build skills, then you demonstrate them through projects, and finally you use that portfolio to land a role.

Foundational technical skills

Foundational machine learning skills are widely considered the most important starting point. This means understanding models such as linear regression, logistic regression, neural networks, decision trees, clustering, and anomaly detection - and, just as importantly, the core concepts behind why machine learning works, including bias and variance, cost functions, regularization, optimization algorithms, and error analysis.

Deep learning has become a large fraction of modern machine learning, so it is valuable to know the basics of neural networks, practical skills such as hyperparameter tuning, and architectures like convolutional networks, sequence models, and transformers.

Software and math

Strong software development skills increase your job opportunities: programming fundamentals, data structures, algorithms, software design, and familiarity with Python. The key Python frameworks to learn are TensorFlow and PyTorch for deep learning, along with scikit-learn for classical machine learning.

The math most relevant to machine learning is linear algebra (vectors and matrices), probability and statistics, and a basic, intuitive understanding of calculus. How much math you need depends on the decisions your work requires; modern tooling means a shallower understanding is often enough to be productive.

Literacy for the AI age

The author compares traditional literacy - reading and writing, which took centuries to spread and enabled deep human-to-human communication - to a new "coding literacy" for AI. Coding for AI is described as the deepest form of human-to-machine communication, and it grows ever more important as machines become more central to daily life.

Habits and continuous learning

Careers are built through consistent learning. A useful idea from BJ Fogg's Tiny Habits is to start small and succeed rather than start big and fail: aspire to study just ten minutes a day, done consistently, and the habit

of daily learning grows naturally over time.